LESSON 8.3
DECOMPOSING QUADRILATERALS TO FIND AREA

LESSON INTRODUCTION

MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.

LEARNING TARGETS

- I can find the area of quadrilaterals by decomposing them into triangles and rectangles.

BENCHMARK COHERENCE

BUILDING ON

MA.5.GR.2.2 Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas.

WORKING TOWARD

MA.7.GR.1.2 Solve mathematical or real-world problems involving the area of polygons or composite figures by decomposing them into triangles or quadrilaterals.

ESSENTIAL QUESTIONS

- How can the decomposition of a figure help you find missing dimensions?
- How can you use the area of triangles and rectangles to find the area of a quadrilateral?
- Is there more than one way to decompose a quadrilateral?

LESSON NARRATIVE

In this lesson, students use what they have learned previously in the unit to decompose quadrilaterals into familiar shapes to determine their area. Students begin by finding the area of a parallelogram by decomposing it into triangles. This strategy is then applied to different quadrilaterals as students decompose them into triangles to find the area. This understanding is extended as students explore the different ways in which shapes can be broken apart to find their area and find the sum of the areas of the smaller shapes to determine the area of the whole.

COMPONENT SUMMARY

8.3.1 WARM-UP *(~5 MIN.)*

Find perimeter and area of a rectangle

8.3.2 COLLABORATION *(10 MIN.)*

Find the area of triangles that make up parallelograms to find the area of a parallelogram

8.3.3 EXPLORATION *(15 MIN.)*

Decompose quadrilaterals to find the area of shapes whose formulas are known

8.3.4 GUIDED PRACTICE *(10 MIN.)*

Decompose quadrilaterals into rectangles and triangles to find their area

8.3.5 YOUR TURN! *(10 MIN.)*

More practice decomposing quadrilaterals in multiple ways into rectangles and triangles to find their area

8.3.6 WRAP-UP *(~5 MIN.)*

Decomposing a quadrilateral to find the area

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Quadrilateral
- Dimensions (length, width, height, base)

MATERIALS

- Separate paper
- Scissors
- Glue
- Four-function calculator

ADDITIONAL PREPARATION

- If you are doing the Collect and Display for 8.3.5 Your Turn! question 3, you may want to display student drawings under the document camera, on their desks, or around the room on the walls.
- Consider precutting the triangles and rectangles to save time.
- If differentiating by content, choose numbers to modify before beginning the lesson.

TEACHER TIPS

COMMON MISCONCEPTIONS

- When decomposing shapes, students may not fully decompose a quadrilateral and get stuck with a shape they do not know how to find the area for.
- Students may forget to solve for missing measurements they need to find the area and use another measurement that does not belong to the decomposed shape.
- Students may incorrectly think that they must have a triangle and rectangle when decomposing.

THINGS TO CONSIDER

- Students need to be reminded that their goal is to break down a quadrilateral into triangles and rectangles.
- Encourage students to find more than one way to decompose the shapes.
- Position students to notice that some quadrilaterals can only be broken down one way, while others will have multiple ways, including different types of shapes used and how many shapes can be used.
- Monitor students and the time it takes them to determine how to decompose the quadrilaterals. If time is a problem, then assign the questions in Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Collect and Display

Consider positioning students to describe a different way they could decompose the shape to a partner while the partner draws and labels the corresponding lengths. Select students to show their drawing to the class and explain how they decomposed their shape. Choose students who decomposed differently to share to give students a different perspective.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

Throughout this lesson, students represent multiple ways to dissect quadrilaterals into different shapes to find their areas. Students explore the ways in which shapes can be partitioned and put back together and connect this to the areas of familiar shapes, such as triangles and rectangles.

DIFFERENTIATE INSTRUCTION

Consider having students preview this lesson by watching the On-Ramp video on classifying quadrilaterals for vocabulary review or finding area for concept review or have the video available to watch for students to opt-in throughout this lesson.

8.3.1 WARM-UP: PERIMETER AND AREA OF A RECTANGLE

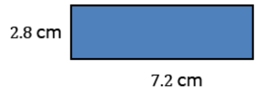
Component Overview: *Students work independently to find the area and perimeter of a rectangle. This activity will help activate prior knowledge of perimeter and area of a rectangle, which will be built upon in this lesson.*

TEACHER MOVES

Position students to write out the formulas for perimeter and area of a rectangle to help with substitution. Activate prior knowledge by prompting students to explain the difference between perimeter and area.

8.3.1 Warm-Up: Perimeter and Area of a Rectangle

1. A rectangle is shown with side lengths given in centimeters (cm).



   a. Find the perimeter of the rectangle. Include units.

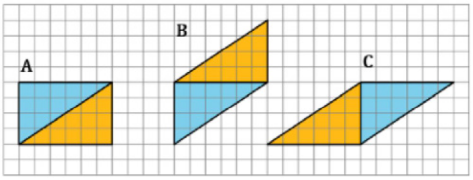
   Perimeter = __20 cm__

   b. Find the area of the rectangle. Include units.

   Area = __20.16 cm^2__

8.3.2 Collaboration: Decomposing Parallelograms

Three parallelograms are shown on the grid. Each square is 1 centimeter (cm) by 1 cm.



1. **Work with your partner to complete the table using the figures shown in the grid. Include units.**

Parallelogram	Area of Blue Triangle	Area of Gold Triangle	Sum of the Two Triangles
A	$12\ cm^2$	$12\ cm^2$	$24\ cm^2$
B	$12\ cm^2$	$12\ cm^2$	$24\ cm^2$
C	$12\ cm^2$	$12\ cm^2$	$24\ cm^2$

2. **Discuss with your partner how the areas of the triangles are related to the area of the parallelogram they form. Write a brief summary of your discussion.**
   Answers will vary.
   The triangle areas can be measured separately and added together to find the area of the parallelogram. If you move the triangles around, the parallelogram still has the same area.

8.3.2 COLLABORATION: DECOMPOSING PARALLELOGRAMS

Component Overview: *Students decompose parallelograms into triangles to find the area of each triangle. Students use the sum of the area of triangles to find the total area of the parallelogram. This positions students to discover how the area of more complex shapes can be found by being broken down into shapes that students already know how to find the area of.*

TEACHER MOVES

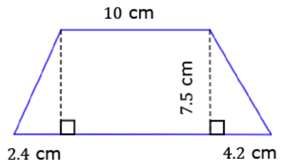
As students work in pairs, circulate to monitor student progress and pose the following questions to deepen understanding and prime students for the rest of the lesson:

- If you rearranged the triangles, would the area still be the same?
- What do you notice about all the areas of the parallelograms?

8.3.3 Exploration: Finding Area by Decomposing

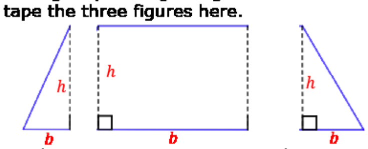
1. **A quadrilateral with measurements in centimeters (cm) is shown.**

Image not drawn to scale



   a. **Copy the image on a piece of paper.**

   b. **Decompose the quadrilateral into two triangles and a rectangle by cutting along the dotted lines. Glue or tape the three figures here.**



   c. **Using the measurements given in the original image, label the base and height of each figure.**

   d. **What is the area of each figure? Include units.**

   Triangle 1: $9\ cm^2$
   Rectangle: $75\ cm^2$
   Triangle 2: $15.75\ cm^2$

   e. **Find the area of the original quadrilateral by finding the sum of the three figures. Include units.**

   $T1 + Rect + T2 = 9cm^2 + 75cm^2 + 15.75cm^2 = 99.75cm^2$

8.3.3 EXPLORATION: FINDING AREA BY DECOMPOSING

Component Overview: *Students will be introduced to finding the area of quadrilaterals by decomposing them into shapes in which they know how to find the area. Students will find the area of each shape and then find their sum, which will be the area of the quadrilateral.*

DIFFERENTIATE INSTRUCTION

After students cut the quadrilateral but before they glue it down, consider giving students time to move the shapes together and apart a few times and letting them rearrange the shapes to see how they could make a quadrilateral in a different arrangement. Prompt students to share what they know about quadrilaterals.

8.3.3 EXPLORATION: FINDING AREA BY DECOMPOSING (CONTINUED)

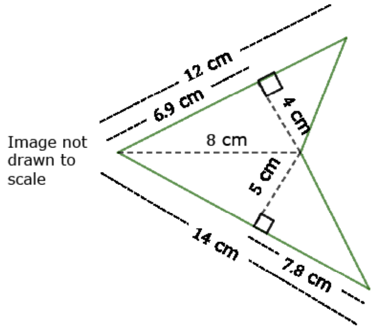
QUESTIONING

On a separate sheet of paper, have students draw three other ways they can decompose the quadrilateral in question 2. Have students share their answers with a partner to compare.

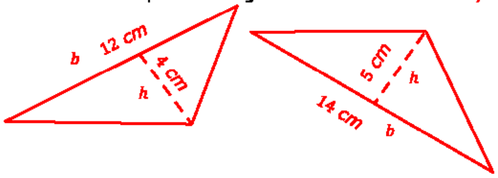
TEACHER MOVES

Overlapping figures are difficult for students to break apart. Ask students to share how they would break apart the quadrilateral. Different pairs may break the quadrilateral up differently. Encourage students to label the length of the sides in question 2a.

2. Another quadrilateral is shown.



a. Decompose the quadrilateral into triangles. Sketch the decomposed triangles here. *Answers will vary.*



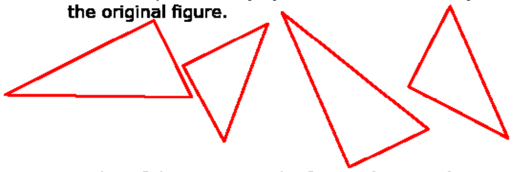
b. Using the measurements given in the original image, label the base and height of each triangle.
Answer shown on triangle sketches.

c. Find the area of the original quadrilateral by finding the sum of the areas of the decomposed triangles. Include units.

$A_1 = \frac{12\ cm \times 4\ cm}{2} = 24\ cm^2$ $A_2 = \frac{14\ cm \times 5\ cm}{2} = 35\ cm^2$

$A_T = A_1 + A_2 = 24\ cm^2 + 35 cm^2 = 59\ cm^2$

d. Sketch any other ways you could have decomposed the original figure.



e. Explain if decomposing the figure changes the area of the original figure.
Answers will vary. Decomposing a figure will not change the original area because you are just breaking apart a figure. You are not expanding or shrinking it.

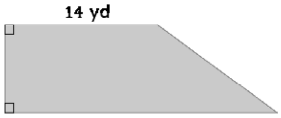
8.3.4 Guided Practice: Finding Area by Decomposing

1. **Find the area of each quadrilateral by decomposing. Lengths are shown in feet (ft). Include units.**

Area = $30\ ft^2$

Image not drawn to scale

2. **The floor plan of a store is shown with measurements in yards (yd).**



   a. **The store found 160 square yards (yd^2) of tile on sale. Based on the floor plan drawn, will the store have enough tiles to redo the flooring if they buy what is on sale?**
   Yes, the area of the floor is $154\ yd^2$.

   b. **Determine the amount of tile, in yd^2, the store is short or has left over.**
   There are $6\ yd^2$ left over.

8.3.4 GUIDED PRACTICE: FINDING AREA BY DECOMPOSING

Component Overview: *Students will continue to practice decomposing quadrilaterals to find their areas. Students will also need to use deductive reasoning and subtraction to find missing side lengths to find the area of the shapes that make up the quadrilaterals.*

TEACHER MOVES

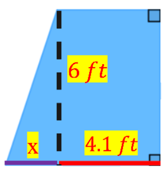
Notice and Wonder

Before students solve each problem, have them decompose the quadrilateral. Then have students participate in a Notice and Wonder routine.

- If you decompose the quadrilateral into a triangle and a rectangle, what do you notice about the measurements for base and height?
- How would you find the base of the triangle with the given base of the quadrilateral and the length of the rectangle?

DIFFERENTIATE INSTRUCTION

Consider having students label the quadrilateral like the example below to visualize how to find the base of the triangle.



8.3.5 YOUR TURN!

Component Overview: *Students will continue to decompose quadrilaterals to find the area. In this activity, students will also need to calculate the missing base of a triangle within the shape using the given information to find the area. Encourage students to mark up and manipulate the quadrilaterals to get a better visual of the decompositions.*

ENRICHMENT

Students may incorrectly think that because there is already a line drawn for the triangle on the right side of question 1 that the quadrilateral will only be decomposed into two shapes. Position students to discuss how this quadrilateral can be further decomposed.

- What are two ways you could figure out that a quadrilateral is not a square or a rectangle?
- Do you have enough information in question 1 to decompose the quadrilateral into two triangles and find their areas?
- Do you have enough information in question 2 to decompose the quadrilateral into two triangles and find their areas?

DIFFERENTIATE INSTRUCTION

Consider having students use manipulatives to create and decompose these shapes. Students could also draw the shapes on paper or create the quadrilaterals using tangrams.

TEACHER MOVES

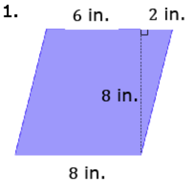
Collect and Display

Consider positioning students to describe a different way they could decompose the shape to a partner while the partner draws and labels the corresponding lengths. Select students to show their drawing to the class and explain how they decomposed their shape. Choose students who decomposed differently to share to give students a different perspective.

8.3.5 Your Turn!

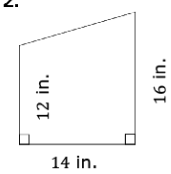
Find the area of each quadrilateral by decomposing. Lengths are shown in inches (in.). Include units.

1.



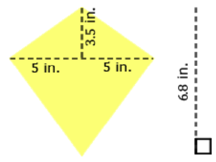
Area = $64\ in^2$

2.



Area = $196\ in^2$

3.

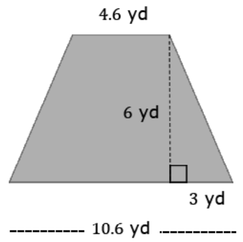


Area = $34\ in^2$

Image not drawn to scale

8.3.6 Wrap-Up: Ticket Out the Door

1. **Find the area of the quadrilateral by decomposing. Lengths are given in yards (yd). Include units.**



Area = $45.6\ yd^2$

8.3.6 WRAP-UP: TICKET OUT THE DOOR

Component Overview: *Students will demonstrate their understanding of decomposing quadrilaterals and finding their areas. Students will need to decompose the quadrilateral into two triangles and a rectangle.*

TEACHER MOVES

Consider requiring students to draw how they decomposed the quadrilateral and to label the decomposition. Their drawing will help determine common misconceptions students may be having and inform your instruction.